

MH2100 Calculus III

Midterm Practice 2 (Questions)

AY 2026/2027, Semester 1

Quantitative Research Society @NTU

Practice material

Overview

- **Coverage:** curves and tangent lines; limits, continuity, partial derivatives and total differentiability; chain rule, directional derivatives, tangent lines, linear approximation and the Implicit Function Theorem; extrema and Lagrange multipliers; order reversal, polar coordinates and volume by a height double integral.
- **Practice design:** four original problems, 100 marks in total; intended for 110 minutes. This practice design is not a confirmed 2026 examination format.
- **Current administration:** Tuesday 6 October 2026, 16:30–18:20, venue TBA. One double-sided typed A4 sheet is permitted; no attachments or sticky notes are permitted. No calculator policy is stated here.

Instructions

Answer **all** questions. Show all working clearly; answers without justification may receive zero marks.

Question 1**(25 Marks)****PRACTICE PROBLEM 1.**

- (a) Let $\mathbf{r}(t) = (e^t, \sin t, t^2)$. Find the point at $t = 0$, a unit tangent vector there, and an equation of the tangent line. **(8 marks)**
- (b) Define $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ by $f(0, 0) = 0$ and $f(x, y) = x^2y/(x^2 + y^2)$ otherwise. Determine continuity at the origin, compute both coordinate partial derivatives there, and decide total differentiability there. Justify each conclusion. **(17 marks)**

Question 2**(25 Marks)****PRACTICE PROBLEM 2.**

- (a) Let h be C^2 near $(0, 1)$, with $h(0, 1) = 2$, $\nabla h(0, 1) = (3, 2)$, and $h_{uu}(0, 1) = 2$, $h_{uv}(0, 1) = 1$, $h_{vv}(0, 1) = -3$. Define $G(x, y) = h(x - y, xy)$. At $P = (1, 1)$, find $G_{xy}(P)$, the directional derivative in the unit direction from P to $(3, 0)$, the tangent line to $G = 2$, and the linear approximation of $G(1.01, 0.98)$. **(15 marks)**
- (b) Let $F(x, y, z) = x + y^2 + e^z - 2$. Use the IFT near $(0, 1, 0)$ to write $z = q(x, y)$. Find $q_x(0, 1)$ and $q_y(0, 1)$, then linearly approximate $q(0.03, 0.99)$. **(10 marks)**

Question 3**(25 Marks)****PRACTICE PROBLEM 3.** Let $f(x, y) = x^2 - y^2$ on $C = \{(x, y) \in \mathbb{R}^2 : x^2 + 9y^2 \leq 9\}$.

- (a) Find and classify every interior critical point. **(8 marks)**
- (b) Use Lagrange multipliers to find every extremum candidate on $x^2 + 9y^2 = 9$. **(12 marks)**
- (c) Use the Extreme Value Theorem to state all absolute extrema on C . **(5 marks)**

Question 4**(25 Marks)****PRACTICE PROBLEM 4.**

- (a) Reverse the order of integration and evaluate

$$\int_0^1 \left[\int_{\sqrt{y}}^1 (1+x) \, dx \right] dy.$$

(10 marks)

- (b) Let D be the part of $x^2 + y^2 \leq 4y$ in the first quadrant. Find the volume above D and below $z = 20 - x^2 - y^2$, using polar coordinates and a height double integral. **(15 marks)**